

Carregando datasets...
Found 3999 files belonging to 5 classes.
Using 3200 files for training.
Found 3999 files belonging to 5 classes.
Using 799 files for validation.
Classes encontradas: ['Other', 'PET', 'PE_HD', 'PP', 'PS']

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TREINANDO MODELO 1: CNN Customizada

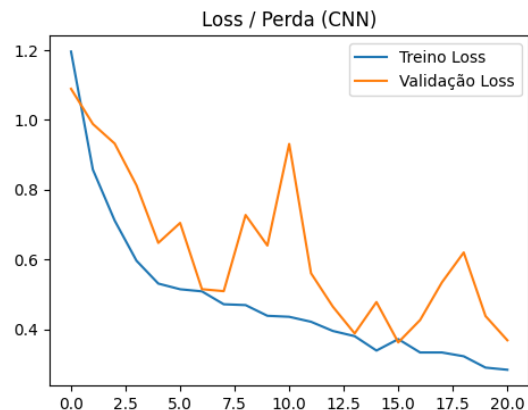
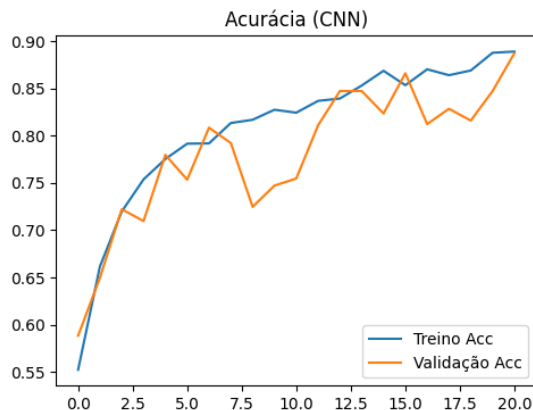
=====

Epoch 1/30
100/100 222s 2s/step -
accuracy: 0.5522 - loss: 1.1957 - val_accuracy: 0.5882 - val_loss: 1.0891
Epoch 2/30
100/100 150s 2s/step -
accuracy: 0.6616 - loss: 0.8576 - val_accuracy: 0.6496 - val_loss: 0.9883
Epoch 3/30
100/100 149s 1s/step -
accuracy: 0.7197 - loss: 0.7117 - val_accuracy: 0.7222 - val_loss: 0.9322
Epoch 4/30
100/100 150s 1s/step -
accuracy: 0.7538 - loss: 0.5966 - val_accuracy: 0.7096 - val_loss: 0.8126
Epoch 5/30
100/100 149s 1s/step -
accuracy: 0.7753 - loss: 0.5311 - val_accuracy: 0.7797 - val_loss: 0.6477
Epoch 6/30
100/100 150s 2s/step -
accuracy: 0.7916 - loss: 0.5149 - val_accuracy: 0.7534 - val_loss: 0.7049
Epoch 7/30
100/100 149s 1s/step -
accuracy: 0.7919 - loss: 0.5088 - val_accuracy: 0.8085 - val_loss: 0.5151
Epoch 8/30
100/100 149s 1s/step -
accuracy: 0.8134 - loss: 0.4719 - val_accuracy: 0.7922 - val_loss: 0.5095
Epoch 9/30
100/100 150s 1s/step -
accuracy: 0.8169 - loss: 0.4695 - val_accuracy: 0.7247 - val_loss: 0.7276
Epoch 10/30
100/100 150s 2s/step -
accuracy: 0.8275 - loss: 0.4389 - val_accuracy: 0.7472 - val_loss: 0.6401
Epoch 11/30

```

100/100          150s 1s/step -
accuracy: 0.8244 - loss: 0.4360 - val_accuracy: 0.7547 - val_loss: 0.9312
Epoch 12/30
100/100          150s 2s/step -
accuracy: 0.8369 - loss: 0.4218 - val_accuracy: 0.8110 - val_loss: 0.5608
Epoch 13/30
100/100          150s 2s/step -
accuracy: 0.8394 - loss: 0.3954 - val_accuracy: 0.8473 - val_loss: 0.4652
Epoch 14/30
100/100          151s 2s/step -
accuracy: 0.8531 - loss: 0.3805 - val_accuracy: 0.8473 - val_loss: 0.3882
Epoch 15/30
100/100          150s 2s/step -
accuracy: 0.8687 - loss: 0.3393 - val_accuracy: 0.8235 - val_loss: 0.4782
Epoch 16/30
100/100          151s 2s/step -
accuracy: 0.8534 - loss: 0.3720 - val_accuracy: 0.8661 - val_loss: 0.3627
Epoch 17/30
100/100          151s 2s/step -
accuracy: 0.8703 - loss: 0.3338 - val_accuracy: 0.8123 - val_loss: 0.4267
Epoch 18/30
100/100          150s 2s/step -
accuracy: 0.8641 - loss: 0.3338 - val_accuracy: 0.8285 - val_loss: 0.5345
Epoch 19/30
100/100          150s 2s/step -
accuracy: 0.8691 - loss: 0.3229 - val_accuracy: 0.8160 - val_loss: 0.6203
Epoch 20/30
100/100          150s 2s/step -
accuracy: 0.8878 - loss: 0.2904 - val_accuracy: 0.8473 - val_loss: 0.4380
Epoch 21/30
100/100          150s 2s/step -
accuracy: 0.8891 - loss: 0.2843 - val_accuracy: 0.8874 - val_loss: 0.3687
Epoch 21: early stopping
Restoring model weights from the end of the best epoch: 16.

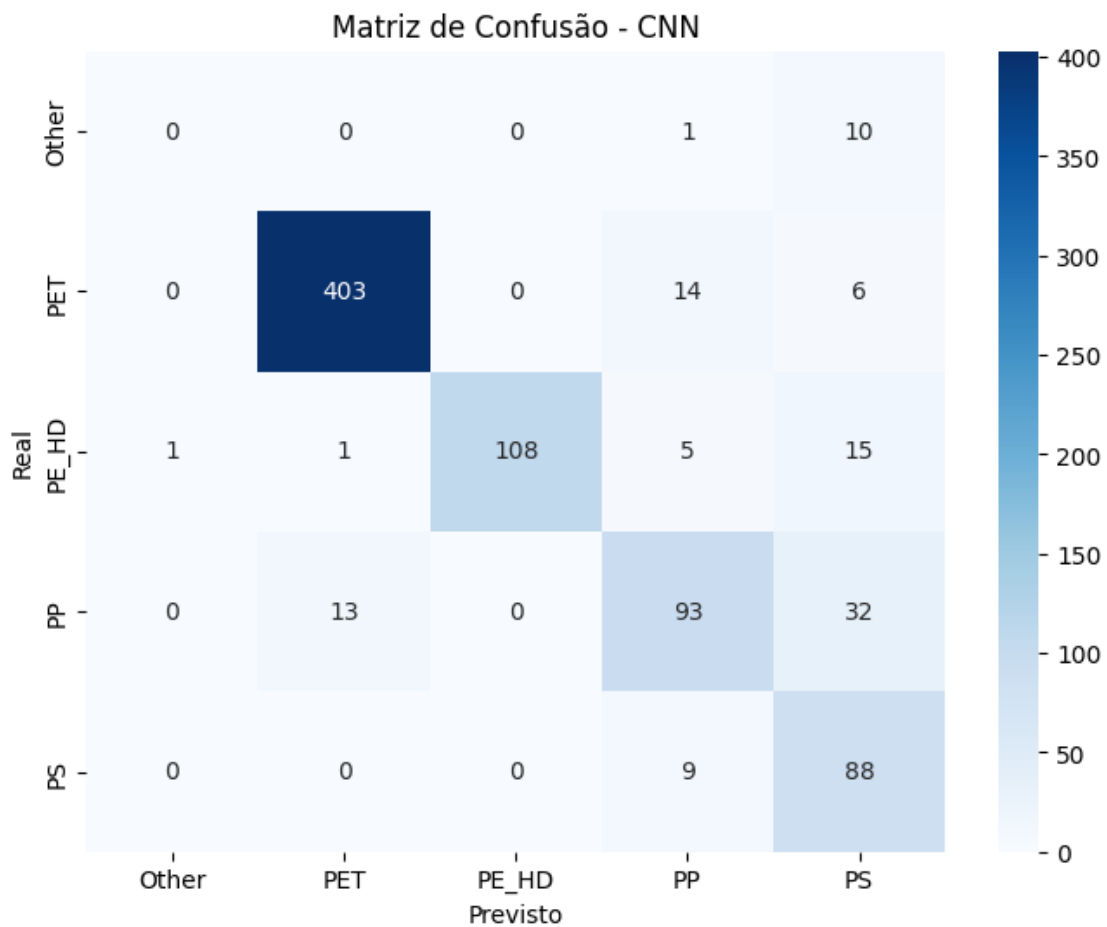
```



--- Avaliação Detalhada: CNN ---

Relatório de Classificação:

	precision	recall	f1-score	support
Other	0.00	0.00	0.00	11
PET	0.97	0.95	0.96	423
PE_HD	1.00	0.83	0.91	130
PP	0.76	0.67	0.72	138
PS	0.58	0.91	0.71	97
accuracy			0.87	799
macro avg	0.66	0.67	0.66	799
weighted avg	0.88	0.87	0.87	799



TREINANDO MODELO 2: Transfer Learning (MobileNetV2)

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h5

9406464/9406464

0s

0us/step

Epoch 1/30

100/100 81s 787ms/step -

accuracy: 0.4412 - loss: 1.5627 - val_accuracy: 0.5494 - val_loss: 1.1807

Epoch 2/30

100/100 78s 778ms/step -

accuracy: 0.5422 - loss: 1.2175 - val_accuracy: 0.5895 - val_loss: 0.9504

Epoch 3/30

100/100 78s 777ms/step -

accuracy: 0.6081 - loss: 1.0247 - val_accuracy: 0.6208 - val_loss: 0.8296

Epoch 4/30

100/100 78s 779ms/step -

accuracy: 0.6500 - loss: 0.9069 - val_accuracy: 0.6796 - val_loss: 0.7265

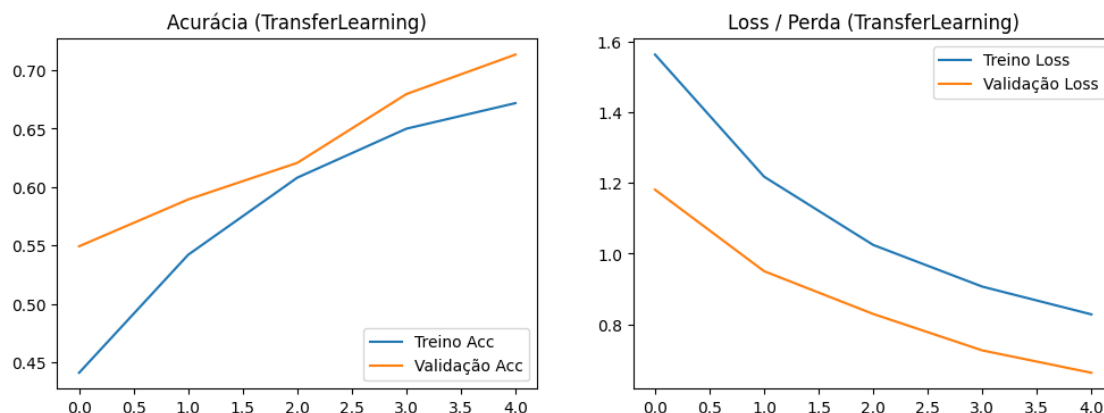
Epoch 5/30

100/100 78s 779ms/step -

accuracy: 0.6719 - loss: 0.8284 - val_accuracy: 0.7134 - val_loss: 0.6633

Epoch 5: early stopping

Restoring model weights from the end of the best epoch: 1.



--- Avaliação Detalhada: TransferLearning ---

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:
 UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels
 with no predicted samples. Use `zero\_division` parameter to control this

behavior.

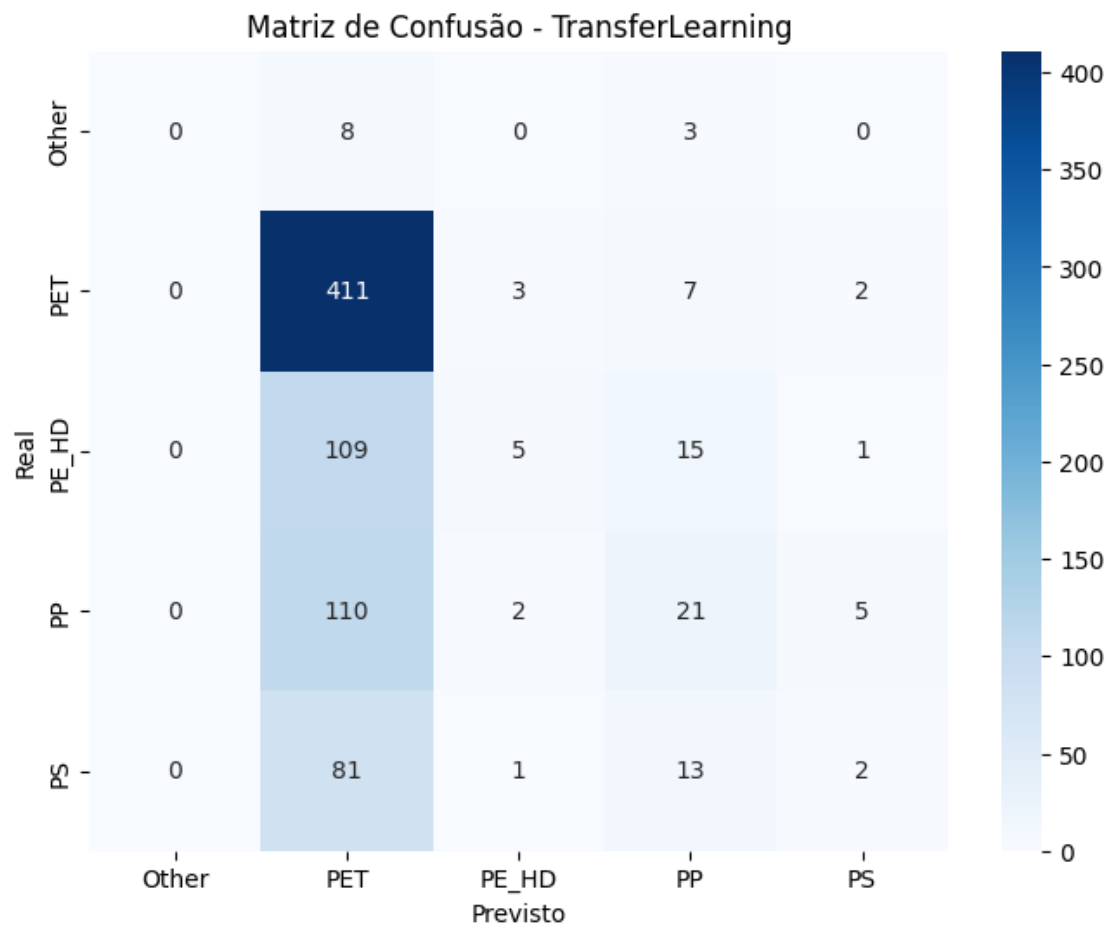
```
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))  
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565:  
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels  
with no predicted samples. Use `zero_division` parameter to control this  
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Relatório de Classificação:

	precision	recall	f1-score	support
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PP	0.36	0.15	0.21	138
PS	0.20	0.02	0.04	97
accuracy			0.55	799
macro avg	0.32	0.24	0.21	799
weighted avg	0.46	0.55	0.43	799



Treinamento Concluído!